

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

- The Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` into a Delta table named "customer".
2. Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" Delta table, calculated as the product of "PricePerUnit" and "Qty".
4. Remove null and duplicate records from both "customer" and "order" Delta tables.

FR-DLT-002: Create and Load Order Summary Table

Title: Create and Load Order Summary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD type 2.

Preconditions:

- The "customer" and "order" Delta tables are loaded with data
- The "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".

2. Join the "customer" and "order" Delta tables on the "CustId" field.
3. Load the joined data into the "ordersummary" table using SCD type 2.

FR-DLT-003: Update Order Summary Table on Customer Changes

Title: Update Order Summary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there is a change in the "customer" table.

Preconditions:

- The "ordersummary" table is loaded with data
- Changes are made to the "customer" table

Main Flow / Functional Steps:

1. Identify changes made to the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" fields accordingly to maintain history.

FR-DLT-004: Create and Load Customer Aggregate Spend Table

Title: Create and Load Customer Aggregate Spend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load it with aggregated data from the "ordersummary" table.

Preconditions:

- The "ordersummary" table is loaded with data
- The "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by "Name" and "Date".
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-005: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements are met

Main Flow / Functional Steps:

1. Implement the loading of customer and order data into Delta tables using Delta Live Tables.
2. Implement the creation and loading of the "ordersummary" table using Delta Live Tables.
3. Implement the update of the "ordersummary" table on customer changes using Delta Live Tables.
4. Implement the creation and loading of the "customeraggregatespend" table using Delta Live Tables.